

Nasal Spray Preparation Guide

Semax 10mg & Selank 10mg — 30ml Nasal Spray

For Research Use Only · Not for Human Consumption

 **DISCLAIMER:** This document is intended for laboratory research purposes only. All procedures described are for in-vitro research use. Not for human consumption. Handle all compounds in a sterile environment using proper PPE.

1. Product Overview



SEMAX

Synthetic analogue of ACTH(4-10). Research applications include neuroprotection, cognitive studies, and stress response modeling.

Molecular Weight: ~858.0 Da
Sequence: Met-Glu-His-Phe-Pro-Gly-Pro
Form: Lyophilized powder



SELANK

Heptapeptide analogue of tuftsin. Studied for anxiolytic effects, immune modulation, and memory enhancement in laboratory settings.

Molecular Weight: ~863.0 Da
Sequence: Thr-Lys-Pro-Arg-Pro-Gly-Pro
Form: Lyophilized powder

2. Materials Required



Peptide Compounds

- Semax 10mg — 1 vial (lyophilized powder)
- Selank 10mg — 1 vial (lyophilized powder)



Reconstitution Solutions

- Bacteriostatic Water (BACwater) — recommended for stability and preservation
- Sterile Water for Injection — alternative if BACwater is unavailable
- Normal Saline (0.9% NaCl) — alternative option
 - Note: BACwater (0.9% Benzyl Alcohol) is preferred as it extends shelf life up to 30 days

Equipment

- 30ml sterile nasal spray bottle (with fine mist atomizer pump)
- 1ml or 3ml sterile syringe
- 18–23 gauge reconstitution needle
- Alcohol wipes / sterile swabs
- Sterile gloves
- Clean work surface or biosafety cabinet

3. Concentration Formula

The following table shows the concentration per pump spray based on the standard 30ml bottle with approximately 0.1ml per pump actuation:

Compound	Concentration	Per Pump Dose
Semax 10mg	10mg / 30ml = 333mcg/ml	~33mcg per pump (0.1ml)
Selank 10mg	10mg / 30ml = 333mcg/ml	~33mcg per pump (0.1ml)
Combined	20mg / 30ml = 667mcg/ml	~67mcg per pump (0.1ml)

 Note: Pump actuation volume varies by bottle type (typically 0.08ml – 0.15ml). Always verify your specific pump volume before use in research protocols.

4. Step-by-Step Preparation

Option A — Semax Only (10mg / 30ml)

1

Prepare your workspace

Wipe down work surface with 70% isopropyl alcohol. Put on sterile gloves. Allow surface to dry completely before proceeding.

2

Measure reconstitution volume

Draw 30ml of Bacteriostatic Water into a sterile syringe. This will be your total volume for the

nasal spray bottle.

3

Reconstitute the Semax vial

Wipe the rubber stopper of the Semax 10mg vial with an alcohol swab. Insert the syringe needle at an angle against the glass wall — do NOT inject directly onto the powder. Slowly inject 1–2ml of BACwater to dissolve the peptide first.

4

Gently mix

Swirl the vial gently in a circular motion for 30–60 seconds. Do NOT shake vigorously as this can denature the peptide. The solution should become completely clear.

5

Transfer to nasal spray bottle

Draw the dissolved solution back into the syringe. Add remaining BACwater directly to the nasal spray bottle first, then add the peptide solution. Total volume = 30ml.

6

Label and seal

Label the bottle with: compound name, concentration (333mcg/ml), preparation date, expiry date (+30 days if using BACwater). Cap the nasal spray bottle securely.

Option B — Selank Only (10mg / 30ml)

Follow the identical steps as Option A, substituting Selank 10mg vial in place of Semax.

Option C — Combined Semax + Selank (10mg each / 30ml)

1

Prepare workspace

As above — clean surface, sterile gloves, alcohol wipes ready.

2

Reconstitute Semax first

Add 1ml BACwater to Semax 10mg vial. Swirl gently until fully dissolved.

3

Reconstitute Selank

Add 1ml BACwater to Selank 10mg vial. Swirl gently until fully dissolved.

4

Combine into spray bottle

Add remaining 28ml BACwater to the nasal spray bottle. Then add both dissolved peptide solutions. Total = 30ml combined solution.

5

Mix and label

Gently invert the bottle 3–5 times to mix. Label with both compound names, combined concentration (667mcg/ml), and preparation date.

5. Storage & Stability

Storage Condition	Shelf Life	Notes
Refrigerator (2–8°C)	Up to 30 days	Recommended storage
Freezer (-20°C)	Up to 3 months	Undiluted powder only
Room temperature	Not recommended	Degrades rapidly

- Always keep away from direct light and heat sources.
- Do NOT freeze reconstituted solution — this may cause peptide aggregation.
- Discard if solution appears cloudy, discolored, or shows particulate matter.

6. Research Tips & Best Practices

Reconstitution Tips

- Always inject BACwater slowly along the glass wall, not directly onto the powder.
- Allow 5–10 minutes for the peptide to dissolve naturally — do not force.
- If powder does not dissolve, gently rotate (do not shake) and wait.

Spray Bottle Tips

- Prime the pump 3–5 times before first use to ensure consistent actuation volume.
- Verify pump volume with water before filling with peptide solution.
- Use opaque or amber-colored bottles to protect from light degradation.

Sterility Tips

- Work in a clean, draft-free environment or use a biosafety cabinet.

- Use a new sterile needle for each vial to prevent cross-contamination.
- Never reuse syringes or needles.

FOR RESEARCH USE ONLY — NOT FOR HUMAN CONSUMPTION

_____ research compounds are intended exclusively for in-vitro laboratory research.

All preparation and handling must comply with applicable research regulations.

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